ANALISI MATEMATICA 2 - FOGLIO 4

1.a $2y-x^2=2+\log(y)$ in $(0,1)$. T_2 di $y=\varphi(x)$ in $x=0$?

Sia $f(x,y)=2y-x^2-2-\log(y)$.

$$f_y(0,1)=\left(2-2x-\frac{1}{y}\right)\Big|_{(0,1)}=1\neq 0 \overset{\text{TFI}}{\Rightarrow} \exists\, y=\varphi(x).$$

$$\varphi(0)=1,\quad \varphi'(0)=-\frac{f_x(0,1)}{f_y(0,1)}=-\frac{(-2x)}{1}\Big|_{(0,1)}=0.$$

Inoltre

$$2\varphi(x)-x^2=2+\log(\varphi(x)) \overset{D}{\Rightarrow} 2\varphi'(x)-2x=\frac{\varphi'(x)}{\varphi(x)}$$

$$\overset{D}{\Rightarrow} 2\varphi''(x)-2=\frac{\varphi''(x)\varphi(x)-(\varphi'(x))^2}{(\varphi(x))}$$

Per $x=0$ otteniamo

$$2\varphi''(0)-2=\frac{\varphi''(0)\cdot 1-0^2}{1^2}\Rightarrow \varphi''(0)=2.$$

Quindi

$$T_2(x)=1+0\cdot x+\frac{2}{2}x^2=1+x^2.$$

1.b $\log(1+x^2-2y)=x+y^4$ in $(0,0)$. T_2 di $y=\varphi(x)$ in $x=0$?

Sia $f(x,y)=\log(1+x^2-2y)-x-y^4$.

$$f_y(0,0)=\left(\frac{-2}{1+x^2-2y}-4y^3\right)\Big|_{(0,0)}=-2\neq 0 \overset{\text{TFI}}{\Rightarrow} \exists\, y=\varphi(x)$$

$$\varphi(0)=0 \text{ e } \varphi'(0)=-\frac{f_x(0,0)}{f_y(0,0)}=-\frac{\left(\frac{2x}{1+x^2-2y}-1\right)|_{(0,0)}}{(-2)}=-\frac{1}{2}.$$

Inoltre

$$\log(1+x^2-2\varphi(x))=x+(\varphi(x))^4 \overset{D}{\Rightarrow} \frac{2x-2\varphi'(x)}{1+x^2-2\varphi(x)}=1+4\varphi^3(x)\varphi'(x)$$

$$\overset{D}{\Rightarrow} \frac{1}{(1+x^2-2\varphi(x))^2}\left[(2-2\varphi''(x))(1+x^2-2\varphi(x))-(2x-2\varphi'(x))^2\right]$$

$$=12\varphi^2(x)(\varphi'(x))^2+4\varphi^3(x)\varphi''(x)$$

Per $x=0$ si ha

$$\frac{1}{1^2}\left[(2-2\varphi''(0))\cdot 1-\left(-2\cdot\left(-\frac{1}{2}\right)\right)^2\right]=0$$

$$2-2\varphi''(0)-1=0 \Rightarrow \varphi''(0)=\frac{1}{2}.$$

Quindi $$T_2(x)=0-\frac{x}{2}+\frac{1}{2}\cdot\frac{1}{2}x^2=-\frac{x}{2}+\frac{x^2}{4}.$$

1.c $y(x^2+y^2)=2x^2$ in $(1,1)$. T_2 di $y=\varphi(x)$ in $x=1$?

Sia $f(x,y)=y(x^2+y^2)-2x^2=yx^2+y^3-2x^2$.

$f_y(1,1)=(x^2+3y^2)|_{(1,1)}=4\neq 0 \overset{\text{TFI}}{\Rightarrow} \exists\, y=\varphi(x)$.

$\varphi(1)=1$, $\varphi'(1)=-\dfrac{f_x(1,1)}{f_y(1,1)}=-\dfrac{(2yx-4x)|_{(1,1)}}{4}=\dfrac{1}{2}$.

Inoltre $\varphi(x)(x^2+\varphi^2(x))=2x^2$

$\overset{D}{\Rightarrow}$ $\varphi'(x)(x^2+\varphi^2(x))+\varphi(x)(2x+2\varphi(x)\varphi'(x))=4x$

$\varphi'(x)x^2+\varphi(x)\cdot 2x+3\varphi^2(x)\varphi'(x)=4x$

$\overset{D}{\Rightarrow}$ $\varphi''(x)x^2+\varphi'(x)\cdot 2x+\varphi'(x)2x+2\varphi(x)$

$+6\varphi(x)(\varphi'(x))^2+3\varphi^2(x)\varphi''(x)=4$

Per $x=1$,

$$\varphi''(1)+\cancel{1}+\cancel{1}+\cancel{2}+6\cdot\frac{1}{4}+3\varphi''(1)=\cancel{4} \Rightarrow \varphi''(1)=-\frac{3}{8}.$$

Quindi $$T_2(x)=1+\frac{1}{2}(x-1)-\frac{3}{16}(x-1)^2.$$

1.d $x^2-2x+4y^2=3$ in $(1,1)$. T_2 di $y=\varphi(x)$ in $x=1$?

Sia $f(x,y)=x^2-2x+4y^2-3$.

$f_y(1,1)=(8y)|_{(1,1)}=8\neq 0 \overset{\text{TFI}}{\Rightarrow} \exists\, y=\varphi(x)$.

$$\varphi(1)=1,\quad \varphi'(1)=-\frac{f_x(1,1)}{f_y(1,1)}=-\frac{(2x-2)|_{(1,1)}}{8}=0.$$

Inoltre $x^2-2x+4\varphi^2(x)=3$

$$\xRightarrow{D} 2x-2+8\varphi(x)\cdot\varphi'(x)=0$$

$$\xRightarrow{D} 2+8\left(\varphi'(x)\right)^2+8\varphi(x)\varphi''(x)=0.$$

Per $x=1$,

$$2+8\cdot 0+8\cdot 1\cdot\varphi''(1)=0 \implies \varphi''(1)=-\frac{1}{4}.$$

Quindi

$$T_2(x)=1-\frac{1}{8}(x-1)^2.$$

1.e $x^2y=e^{xy+1}$ in $(-1,1)$. T_2 di $y=\varphi(x)$ in $x=-1$?

Sia $f(x,y)=x^2y-e^{xy+1}$.

$$f_y(-1,1)=\left(x^2-e^{xy+1}\cdot x\right)|_{(-1,1)}=2\neq 0 \xRightarrow{TFI} \exists\, y=\varphi(x).$$

$$\varphi(-1)=1,\quad \varphi'(-1)=-\frac{f_x(-1,1)}{f_y(-1,1)}=-\frac{(2xy-e^{xy+1}y)|_{(-1,1)}}{2}=\frac{3}{2}.$$

Inoltre $x^2\varphi(x)=e^{x\varphi(x)+1}$

$$\xRightarrow{D} 2x\varphi(x)+x^2\varphi'(x)=e^{x\varphi(x)+1}\left(\varphi(x)+x\varphi'(x)\right)$$

$$\xRightarrow{D} 2\varphi(x)+4x\varphi'(x)+x^2\varphi''(x)$$

$$=e^{x\varphi(x)+1}\left[\left(\varphi(x)+x\varphi'(x)\right)^2+\left(2\varphi'(x)+x\varphi''(x)\right)\right]$$

Per $x=-1$,

$$2\cdot 1+4(-1)\cdot\frac{3}{2}+1\cdot\varphi''(-1)=1\left[\left(1-\frac{3}{2}\right)^2+\left(2\cdot\frac{3}{2}-1\cdot\varphi''(-1)\right)\right]$$

da cui

$$\varphi''(-1)=\frac{1}{2}\left(\frac{1}{4}+3-2+6\right)=\frac{29}{8}.$$

Quindi

$$T_2(x)=1+\frac{3}{2}(x+1)+\frac{29}{16}(x+1)^2.$$

1.f $x^5+3x^2y-2y^4=1$ in $(1,0)$. T_2 di $y=\varphi(x)$ in $x=1$?

Sia $f(x,y)=x^5+3x^2y-2y^4-1$.

$$f_y(1,0)=(3x^2-8y^3)_{|(1,0)}=3\neq 0 \overset{\text{TFI}}{\Rightarrow} \exists\, y=\varphi(x).$$

$$\varphi(1)=0,\quad \varphi'(1)=-\frac{f_x(1,0)}{f_y(1,0)}=-\frac{(5x^4+6xy)}{3}_{|(1,0)}=-\frac{5}{3}.$$

Inoltre $x^5+3x^2\varphi(x)-2\varphi^4(x)=1$

$$\overset{D}{\Rightarrow} 5x^4+6x\varphi(x)+3x^2\varphi'(x)-8\varphi^3(x)\cdot\varphi'(x)=0$$

$$\overset{D}{\Rightarrow} 20x^3+6\varphi(x)+12x\varphi'(x)+3x^2\varphi''(x)$$
$$-24\varphi^2(x)\cdot\varphi'(x)-8\varphi^3(x)\varphi''(x)=0$$

Per $x=1$,

$$20+0+12\left(-\frac{5}{3}\right)+3\varphi''(1)=0 \Rightarrow \varphi''(1)=0.$$

Quindi

$$T_2(x)=-\frac{5}{3}(x-1).$$

2.a $\begin{cases} xyz=-2 \\ x^3+y^3+z^3=8 \end{cases}$ in $(-1,2,1)$. T_1 di $y=\varphi(x)$ e $z=\psi(x)$ in $x=-1$?

Siano $f(x,y,z)=xyz+2=0$, $g(x,y,z)=x^3+y^3+z^3-8=0$.

$$J=\begin{bmatrix} f_y & f_z \\ g_y & g_z \end{bmatrix}=\begin{bmatrix} xz & xy \\ 3y^2 & 3z^2 \end{bmatrix} \overset{(-1,2,1)}{\Longrightarrow} \begin{bmatrix} -1 & -2 \\ 12 & 3 \end{bmatrix} \Longrightarrow \det(J)=21\neq 0$$

Quindi per TFI $\exists\, y=\varphi(x)$ e $\exists\, z=\psi(x)$: $\varphi(-1)=2$, $\psi(-1)=1$.
Inoltre derivando

$$\begin{cases} x\varphi(x)\psi(x)=-2 \\ x^3+(\varphi(x))^3+(\psi(x))^3=8 \end{cases}$$

si ottiene

$$\begin{cases} \varphi(x)\psi(x)+x\varphi'(x)\psi(x)+x\varphi(x)\psi'(x)=0 \\ 3x^2+3(\varphi(x))^2\cdot\varphi'(x)+3(\psi(x))^2\cdot\psi'(x)=0 \end{cases}$$

Per $x=-1$ si ha

$$\begin{cases} 2\cdot 1-\varphi'(-1)\cdot 1-2\psi'(-1)=0 \\ 3+12\varphi'(-1)+3\psi'(-1)=0 \end{cases} \qquad \begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix}\begin{bmatrix} \varphi'(-1) \\ \psi'(-1) \end{bmatrix}=\begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

da cui

$$\varphi'(-1)=\frac{\begin{vmatrix} 2 & 2 \\ -1 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & 2 \\ 4 & 1 \end{vmatrix}}=\frac{4}{-7} \quad \text{e} \quad \psi'(-1)=\frac{\begin{vmatrix} 1 & 2 \\ 4 & -1 \end{vmatrix}}{\begin{vmatrix} 1 & 2 \\ 4 & 1 \end{vmatrix}}=\frac{-9}{-7}=\frac{9}{7}.$$

REGOLA DI CRAMER

Infine

$$\varphi)\ T_1(x)=2-\frac{4}{7}(x+1) \qquad \psi)\ T_1(x)=1+\frac{9}{7}(x+1).$$

2.b $\begin{cases} x^2+y^2-z^2=1 \\ xy+yz=xz-3 \end{cases}$ in $(1,-1,1)$. T_1 di $y=\varphi(x)$ e $z=\psi(x)$ in $x=1$?

Siamo

$$f(x,y,z)=x^2+y^2-z^2-1=0, \quad g(x,y,z)=xy+yz-xz+3=0$$

$$J=\begin{bmatrix} f_y & f_z \\ g_y & g_z \end{bmatrix}=\begin{bmatrix} 2y & -2z \\ x+z & y-x \end{bmatrix} \overset{(1,-1,1)}{\Longrightarrow} \begin{bmatrix} -2 & -2 \\ 2 & -2 \end{bmatrix} \Longrightarrow \det(J)=16\neq 0$$

Quindi per TFI $\exists\, y=\varphi(x)$ e $\exists\, z=\psi(x)$: $\varphi(1)=-1$ e $\psi(1)=1$.

Inoltre derivando

$$\begin{cases} x^2+\varphi^2(x)-\psi^2(x)=1 \\ x\varphi(x)+\varphi(x)\psi(x)=x\psi(x)-3 \end{cases}$$

si ottiene

$$\begin{cases} 2x+2\varphi(x)\varphi'(x)-2\psi(x)\psi'(x)=0 \\ \varphi(x)+x\varphi'(x)+\varphi'(x)\psi(x)+\varphi(x)\psi'(x)=\psi(x)+x\psi'(x) \end{cases}$$

Per $x=1$ si ha

$$\begin{cases} 2-2\varphi'(1)-2\psi'(1)=0 \\ -1+\varphi'(1)+\varphi'(1)-\psi'(1)=1+\psi'(1) \end{cases} \qquad \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \varphi'(1) \\ \psi'(1) \end{bmatrix}=\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

da cui

$$\varphi'(1)=\frac{\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}}{\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}}=1 \qquad \psi'(1)=\frac{\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}}=0$$

Infine

$\varphi)\ T_1(x)=-1+(x-1)=x-2$, $\psi)\ T_1(x)=1+0\cdot(x-1)=1$.

3.a $\{(x,y,z) : xyz=-8\}$ Piano tangente in $(-2,1,4)$?

Sia $f(x,y,z)=xyz+8$. Allora

$$\nabla f(x,y,z)=(yz,xz,xy)$$

e il piano tangente in $(-2,1,4)$ è

$$f_x(-2,1,4)(x+2)+f_y(-2,1,4)(y-1)+f_z(-2,1,4)(z-4)=0$$

ossia $4(x+2)-8(y-1)-2(z-4)=0$

$$2x-4y-z+12=0.$$

3.b $\{(x,y,z):(x-1)^2+(y-2)^2+4(z-2)^2=9\}$

Piano tangente in $(3,1,3)$?

Sia $f(x,y,z)=(x-1)^2+(y-2)^2+4(z-2)^2-9$. Allora

$$\nabla f(x,y,z)=(2(x-1),2(y-2),8(z-2))$$

e dunque $\nabla f(3,1,3)=(4,-2,8)$.

Così il piano tangente in $(3,1,3)$ è

$$\cancel{4}^{2}(x-3)-\cancel{2}^{1}(y-1)+\cancel{8}^{4}(z-3)=0$$

ossia

$$2x-y+4z=17.$$

3.c $\{(x,y,z):(x^2+y^2-z^2)(x^2+y^2-8z^2)=0\}$

Piano tangente in $(2,2,1)$?

Sia $f(x,y,z)=(x^2+y^2-z^2)(x^2+y^2-8z^2)$. Allora

$$\nabla f(x,y,z)=(2x,2y,-2z)(x^2+y^2-8z^2)+(x^2+y^2-z^2)(2x,2y,-16z)$$

e quindi $\nabla f(2,2,1)=(4,4,-2)\cdot 0+7\cdot(4,4,-16)=28(1,1,-4)$.

Così il piano richiesto è

$$1\cdot(x-2)+1\cdot(y-2)-4(z-1)=0 \Rightarrow x+y-4z=0.$$

4.a $f(x,y)=xy+x^2$ Punti stazionari vincolati in $\Gamma=\{(x,y): x^2+y^2+xy=4\}$?

Sia $g(x,y)=x^2+y^2+xy-4=0$ allora

$\nabla g(x,y)=(2x+y,2y+x)=(0,0) \iff (x,y)=(0,0)\notin\Gamma$

Tutti i punti di Γ sono regolari

Punti stazionari vincolati

$$\begin{cases}\nabla f(x,y)=\lambda\nabla g(x,y)\\ g(x,y)=0\end{cases} \quad \begin{cases}y+2x=\lambda(2x+y)\\ x=\lambda(2y+x)\\ x^2+y^2+xy=4\end{cases} \to (\lambda-1)(2x+y)=0$$

$\lambda=1$ $\quad y=-2x$

$$\begin{cases}\lambda=1\\ x=2y+x \to y=0\\ x^2+y^2+xy=4 \to x^2=4 \to x=\pm 2\end{cases} \Rightarrow \boxed{\begin{matrix}(2,0)\\(-2,0)\end{matrix}} \quad \lambda=1$$

$$\begin{cases}y=-2x \to y=\mp\frac{4}{\sqrt{3}}\\ x=\lambda(-4x+x) \to \lambda=-\frac{1}{3}\\ x^2+4x^2-2x^2=4 \to x=\pm\frac{2}{\sqrt{3}}\\ 3x^2=4\end{cases} \Rightarrow \boxed{\begin{matrix}\left(\frac{2}{\sqrt{3}},-\frac{4}{\sqrt{3}}\right)\\ \left(-\frac{2}{\sqrt{3}},\frac{4}{\sqrt{3}}\right)\end{matrix}} \quad \lambda=-\frac{1}{3}$$

4.b $f(x,y)=(3x+2y)^2$ Punti stazionari vincolati in $\Gamma=\{(x,y): 4x^2+y^2=4\}$?

Sia $g(x,y)=4x^2+y^2-4=0$ allora

$\nabla g(x,y)=(8x,2y)=(0,0) \iff (x,y)=(0,0)\notin\Gamma$

Tutti i punti di Γ sono regolari

Punti stazionari vincolati:

$$\begin{cases}\nabla f(x,y)=\lambda\nabla g(x,y)\\ g(x,y)=0\end{cases} \quad \begin{cases}2(3x+2y)\cdot 3=\lambda(8x)\\ 2(3x+2y)\cdot 2=\lambda(2y)\\ 4x^2+y^2=4\end{cases} \quad 2\textcircled{1}-3\textcircled{2}$$

$$\begin{cases} 8\lambda x - 3\lambda y = 0 \\ (3x+2y)\cdot 2 = \lambda y \\ 4x^2+y^2=4 \end{cases} \rightarrow \lambda(8x-3y)=0 \quad \begin{matrix} \lambda=0 \\ y=\frac{8}{3}x \end{matrix}$$

$$\begin{cases} \lambda = 0 \\ 3x+2y=0 \\ 4x^2+y^2=4 \end{cases} \rightarrow \begin{matrix} \\ y=-\frac{3x}{2} \\ 4x^2+\frac{9x^2}{4}=4 \end{matrix} \rightarrow \begin{matrix} \\ y=\mp\frac{6}{5} \\ x=\pm\frac{4}{5} \end{matrix} \qquad \boxed{\begin{matrix} \left(\frac{4}{5},-\frac{6}{5}\right) \\ \left(-\frac{4}{5},\frac{6}{5}\right) \end{matrix}} \quad \lambda=0$$

$$\begin{cases} y=\frac{8}{3}x \\ (3x+2(\frac{8}{3}x))\cdot 2 = \lambda(\frac{8}{3}x) \\ 4x^2+\frac{64}{9}x^2=4 \end{cases} \begin{matrix} \rightarrow y=\pm\frac{8}{5} \\ \rightarrow \lambda=\frac{25}{4} \\ \rightarrow x=\pm\frac{3}{5} \end{matrix} \qquad \boxed{\begin{matrix} \left(\frac{3}{5},\frac{8}{5}\right) \\ \left(-\frac{3}{5},-\frac{8}{5}\right) \end{matrix}} \quad \lambda=\frac{25}{4}$$

4.c $f(x,y)=x-x^2-y^2$ Punti stazionari vincolati in $\Gamma=\{(x,y): x^2+2y^2=9\}$?

Sia $g(x,y)=x^2+2y^2-9=0$ allora

$$\nabla g(x,y)=(2x,4y)=(0,0) \iff (x,y)=(0,0)\notin\Gamma$$

Tutti i punti di Γ sono regolari.

Punti stazionari vincolati:

$$\begin{cases} \nabla f(x,y)=\lambda\nabla g(x,y) \\ g(x,y)=0 \end{cases} \qquad \begin{cases} 1-2x=\lambda 2x \\ -2y=\lambda 4y \\ x^2+2y^2=9 \end{cases} \rightarrow y(2\lambda+1)=0 \quad \begin{matrix} y=0 \\ \lambda=-\frac{1}{2} \end{matrix}$$

$$\begin{cases} y=0 \\ 1-2x=\lambda 2x \\ x^2=9 \end{cases} \quad \cup \quad \begin{cases} \lambda=-\frac{1}{2} \\ 1-2x=-x \rightarrow x=1 \\ x^2+2y^2=9 \rightarrow y=\pm 2 \end{cases}$$

$x=3$ $\lambda=-\frac{5}{6}$ $\quad x=-3$ $\lambda=-\frac{7}{6}$

$\boxed{(3,0),(-3,0)}$ $\qquad \boxed{(1,2),(1,-2)}$

4.d $f(x,y)=\log(xy)$ Punti stazionari vincolati in $\Gamma=\{(x,y): x^2+2y^2=y\}$?

$D=\{(x,y): xy>0\}$.

Sia $g(x,y)=x^2+2y^2-y=0$ allora

$$\nabla g(x,y)=(2x,4y-1)=(0,0) \iff (x,y)=(0,\tfrac{1}{4})\notin\Gamma$$

Tutti i punti di Γ sono regolari.

Punti stazionari vincolati:

$$\begin{cases}\nabla f(x,y)=\lambda\nabla g(x,y)\\ g(x,y)=0\\ xy>0\end{cases}\qquad \begin{cases}\frac{1}{x}=\lambda 2x \to x^2=\frac{1}{2\lambda}\\ \frac{1}{y}=\lambda(4y-1)\to 4y^2-y=2x^2\\ x^2+2y^2=y\to 2y^2-\frac{y}{2}+2y^2=y\\ 4y^2=\frac{3}{2}y,\ y\neq 0\\ \to y=\frac{3}{8}\end{cases}$$

$$\begin{cases}y=\frac{3}{8}\\ 2x^2=4\cdot\frac{9}{64}-\frac{3}{8}=\frac{3}{16}\to x=\pm\frac{\sqrt{6}}{8}\\ \lambda=\frac{16}{3}\end{cases}$$

$$\boxed{\left(\tfrac{\sqrt{6}}{8},\tfrac{3}{8}\right)}$$

4.e $f(x,y)=x^2+(y-5)^2$ Punti stazionari vincolati in $\Gamma=\{(x,y): y^2+1=x\}$?

Sia $g(x,y)=y^2+1-x=0$ allora

$$\nabla g(x,y)=(-1,2y)=(0,0)\quad \text{impossibile}$$

Tutti i punti di Γ sono regolari.

Punti stazionari vincolati:

$$\begin{cases}\nabla f(x,y)=\lambda\nabla g(x,y)\\ g(x,y)=0\end{cases}\qquad \begin{cases}2x=-\lambda\\ 2(y-5)=2\lambda y\to y-5=-2xy\to y\neq 0\\ y^2+1=x\end{cases}$$

$\cdot 2y$: $2y(y^2+1)=2xy=-y+5$

$2y^3+3y-5=0$

$(y-1)(2y^2+2y+5)=0$, $\Delta<0$

da cui $y=1$ e $x=1^2+1=2$

$$\boxed{(2,1)}$$

4.F $f(x,y)=2x+y$ Punti stazionari vincolati in $\Gamma=\{(x,y):(x+1)^2+y^2=5\}$?

Sia $g(x,y)=(x+1)^2+y^2-5=0$ allora

$$\nabla g(x,y)=(2(x+1),2y)=(0,0) \iff (x,y)=(-1,0)\notin\Gamma$$

Tutti i punti di Γ sono regolari.

Punti stazionari vincolati:

$$\begin{cases}\nabla f(x,y)=\lambda\nabla g(x,y)\\ g(x,y)=0\end{cases} \quad \begin{cases}2=\lambda 2(x+1) \to x+1=\frac{1}{\lambda}\\ 1=\lambda 2y \to y=\frac{1}{2\lambda}\\ (x+1)^2+y^2=5 \to \frac{1}{\lambda^2}+\frac{1}{4\lambda^2}=5\end{cases}$$

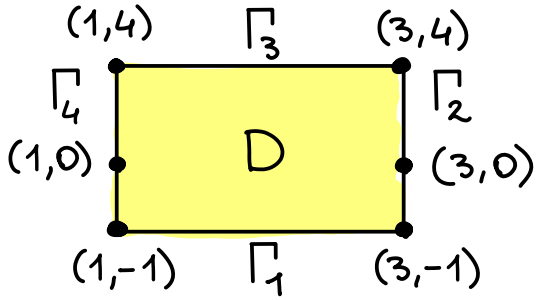
$\lambda=\pm\frac{1}{2}$

$$\begin{cases}\lambda=\frac{1}{2}\\ x=2-1=1\\ y=1\end{cases} \qquad \begin{cases}\lambda=-\frac{1}{2}\\ x=-2-1=-3\\ y=-1\end{cases}$$

$\boxed{(1,1),(-3,-1)}$

5.a $f(x,y)=\log((x+1)^2+y^2)$ Max/min in $D=[1,3]\times[-1,4]$?

$f\in C^2(\mathbb{R}^2\setminus\{(-1,0)\})$, D è compatto e $(-1,0)\notin D$.



$\partial D=\Gamma_1\cup\Gamma_2\cup\Gamma_3\cup\Gamma_4$

Punti stazionari interni a D:

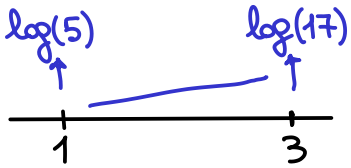
$$\nabla f(x,y)=\left(\frac{2(x+1)}{(x+1)^2+y^2},\frac{2y}{(x+1)^2+y^2}\right)=(0,0)$$

Senza soluzioni in $\mathbb{R}^2\setminus\{(-1,0)\}$

Bordo di D: $\partial D=\Gamma_1\cup\Gamma_2\cup\Gamma_3\cup\Gamma_4$

1) $\Gamma_1=\{(x,-1):x\in[1,3]\}$,

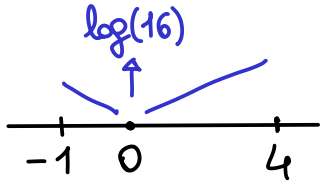
$h(x)=f(x,-1)=\log((x+1)^2+1)$



$f(1,-1)=\log(5)$, $f(3,-1)=\log(17)$

2) $\Gamma_2=\{(3,y):y\in(-1,4)\}$,

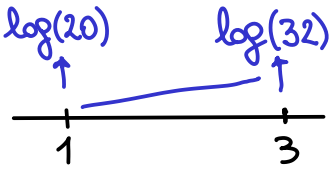
$h(y)=f(3,y)=\log(16+y^2)$



$f(3,0)=\log(16)$

3) $\Gamma_3=\{(x,4):x\in[1,3]\}$,

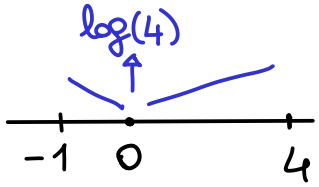
$h(x)=f(x,4)=\log((x+1)^2+16)$



$f(1,4)=\log(20)$, $f(3,4)=\log(32)$

4) $\Gamma_4 = \{(1,y) : y \in (-1,4)\}$,

$h(y) = f(1,y) = \log(4+y^2)$

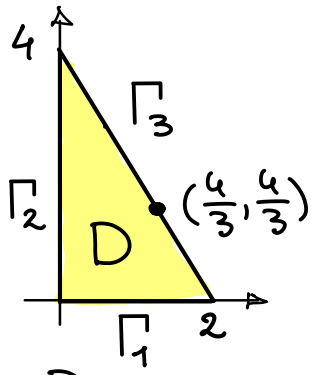


$$\boxed{f(1,0) = \log(4)}$$

Confrontando i valori trovati si ha che $(1,0)$ è un punto di minimo assoluto con valore $\log(4)$ e $(3,4)$ è un punto di massimo assoluto con valore $\log(32)$.

5.6 $f(x,y) = x^2 y$ Max/min in $D = \{(x,y) : 2x+y \leq 4, x \geq 0, y \geq 0\}$?

$f \in C^2(\mathbb{R}^2)$, D è compatto.



Punti stazionari interni a D:

$\nabla f(x,y) = (2xy, x^2) = (0,0)$

$x = 0$ e $\forall y \in \mathbb{R}$.

Nessun punto stazionario è interno a D.

Bordo di D: $\partial D = \Gamma_1 \cup \Gamma_2 \cup \Gamma_3$

1) $\Gamma_1 = \{(x,0) : x \in [0,2]\}$

$h(x) = \boxed{f(x,0) = 0}$ costante

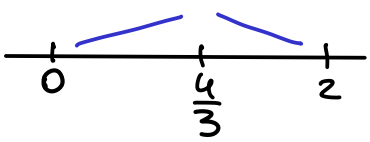
2) $\Gamma_2 = \{(0,y) : y \in [0,4]\}$

$h(x) = \boxed{f(0,y) = 0}$ costante

3) $\Gamma_3 = \{(x, 4-2x) : x \in (0,2)\}$

$h(x) = f(x, 4-2x) = x^2 \cdot (4-2x) = 4x^2 - 2x^3$

$h'(x) = 8x - 6x^2 = 2x(4-3x)$

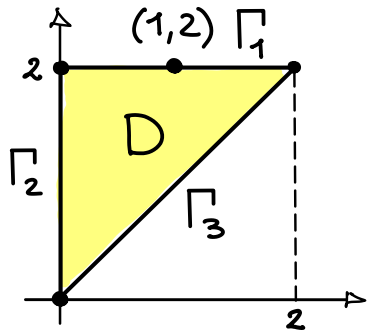


$$f\left(\frac{4}{3},\frac{4}{3}\right) = \left(\frac{4}{3}\right)^3 = \frac{64}{27}$$

Confrontando i valori trovati si ha che i punti di $\Gamma_1 \cup \Gamma_2$ sono punti di minimo assoluto con valore 0 e $\left(\frac{4}{3},\frac{4}{3}\right)$ è un punto di massimo assoluto con valore $\frac{64}{27}$.

5.C $f(x,y) = x^2 + 2y^2 - xy$ Max/min in $D = \{(x,y) : 0 \le x \le y \le 2\}$?

$f \in C^2(\mathbb{R}^2)$, D è compatto.



Punti stazionari interni a D:

$$\nabla f(x,y) = (2x-y, 4y-x) = (0,0)$$

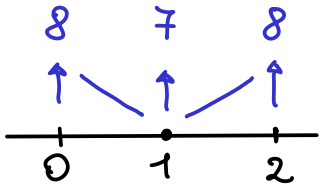
da cui si ottiene (0,0) che non è interno a D.

Bordo di D: $\partial D = \Gamma_1 \cup \Gamma_2 \cup \Gamma_3$

1) $\Gamma_1 = \{(x,2) : x \in [0,2]\}$

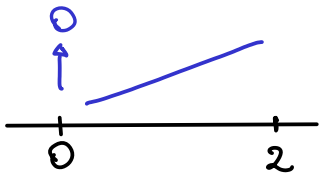
$h(x) = f(x,2) = x^2 + 8 - 2x$

$h'(x) = 2x - 2$



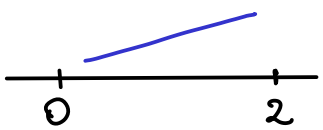
2) $\Gamma_2 = \{(0,y) : y \in [0,2)\}$

$h(y) = f(0,y) = 2y^2$



3) $\Gamma_3 = \{(x,x) : x \in (0,2)\}$

$h(x) = f(x,x) = 2x^2$

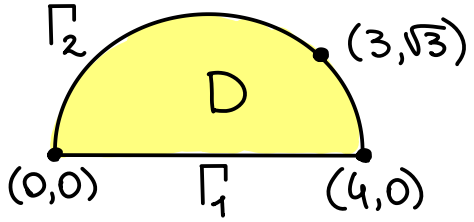


Confrontando i valori trovati si ha che $(2,2)$ e $(0,2)$ sono punti di massimo assoluto con valore 8 e $(0,0)$ è un punto di minimo assoluto con valore 0.

5.d $f(x,y)=x+\sqrt{3}y$ Max/min in $D=\{(x,y): x^2+y^2\le 4x, y\ge 0\}$?

$\hookrightarrow (x-2)^2+y^2\le 4$

$f\in C^2(\mathbb{R}^2)$ e D è compatto



$\partial D=\Gamma_1\cup\Gamma_2$

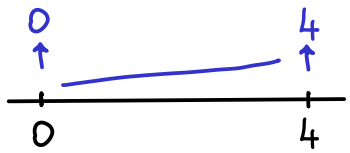
Non ci sono punti stazionari interni perché

$$\nabla f(x,y)=(1,\sqrt{3})\neq(0,0)$$

Bordo di D: $\partial D=\Gamma_1\cup\Gamma_2$ $\boxed{f(0,0)=0,\ f(4,0)=4}$

1) $\Gamma_1=\{(x,0): x\in[0,4]\}$

$h(x)=f(x,0)=x$



2) $\Gamma_2=\{(x,y): \underbrace{x^2+y^2-4x}_{=g(x,y)}=0, y>0\}$

$\nabla g(x,y)=(2x-4,2y)=(0,0)\Rightarrow (x,y)=(2,0)\notin\Gamma_2$ tutti regolari

Punti stazionari vincolati: $\begin{cases}\nabla f=\lambda\nabla g\\ g=0\end{cases}$

$$\begin{cases}1=\lambda(2x-4)\\ \sqrt{3}=\lambda(2y)\\ x^2+y^2=4x\end{cases}$$

$\to y=\sqrt{3}(x-2)$

$\to \lambda=\frac{\sqrt{3}}{2y}$

$\to x^2+3(x-2)^2=4x,\ x^2+3x^2-12x+12=4x$

$x^2-4x+3=0\to x=2\pm 1=\begin{cases}1\\ 3\end{cases}$

$$\begin{cases} x=1 \\ y=-\sqrt{3} \\ \lambda=-\frac{1}{2} \end{cases} \cup \begin{cases} x=3 \\ y=\sqrt{3} \\ \lambda=\frac{1}{2} \end{cases} \qquad (1,-\sqrt{3}) \notin \Gamma_2 \qquad (3,\sqrt{3}) \in \Gamma_2 \qquad \boxed{f(3,\sqrt{3})=6}$$

Confrontando i valori trovati si ha che $(0,0)$ è un punto di minimo assoluto con valore 0 e $(3,\sqrt{3})$ è un punto di massimo assoluto con valore 6.

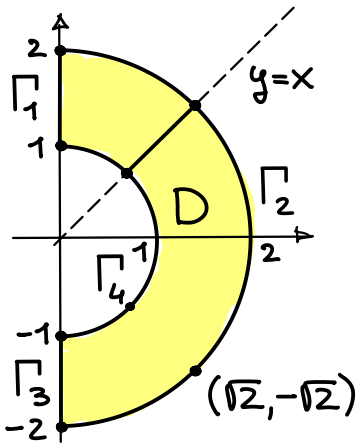
5. e $f(x,y)=(y-x)^2$ Max/min in $D=\{(x,y): 1\le x^2+y^2\le 4, x\ge 0\}$?

$f\in C^2(\mathbb{R}^2)$, D è compatto.

Punti stazionari interni a D:

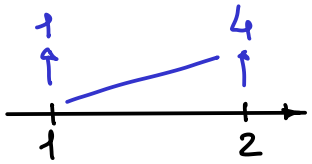
$$\nabla f(x,y)=(2x-2y,2y-2x)=(0,0)$$

da cui $y=x$. Quindi i punti stazionari interni a D sono $(x,x): x\in(\frac{1}{\sqrt{2}},\sqrt{2})$ e $\boxed{f(x,x)=0}$. Dato che $f\ge 0$ in $\mathbb{R}^2$ tali punti sono di minimo assoluto.



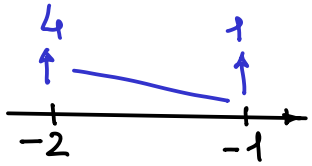
Bordo di D: $\partial D=\Gamma_1\cup\Gamma_2\cup\Gamma_3\cup\Gamma_4$

1) $\Gamma_1=\{(0,y): y\in[1,2]\}$ $f(0,y)=y^2$



con $\boxed{f(0,1)=1}$ e $\boxed{f(0,2)=4}$.

$\Gamma_3=\{(0,y): y\in[-2,-1]\}$ $f(0,y)=y^2$



con $\boxed{f(0,-2)=4}$ e $\boxed{f(0,-1)=1}$.

2) $\Gamma_2=\{(x,y): x^2+y^2=4, x>0\}$, $\Gamma_4=\{(x,y): x^2+y^2=1, x>0\}$

Sia $g(x,y)=x^2+y^2-r^2=0$ con $r=1,2$.

$\nabla g(x,y)=(2x,2y)\neq(0,0)$ per $(x,y)\in\Gamma_2\cup\Gamma_4$ tutti regolari

Punti stazionari vincolati:

$$\begin{cases} 2x-2y=\lambda 2x \\ 2y-2x=\lambda 2y \\ x^2+y^2=r^2 \end{cases}$$

$\cdot y \to xy-y^2=\lambda xy$

$\cdot x \to xy-x^2=\lambda xy$

↓ sottraendo

$x^2=y^2$

$$\begin{cases} x=\pm y \\ 2x^2=r^2 \end{cases} \quad x>0 \Rightarrow \left(\frac{r}{\sqrt2},\frac{r}{\sqrt2}\right), \left(\frac{r}{\sqrt2},-\frac{r}{\sqrt2}\right)$$

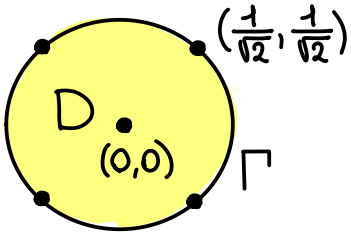
Per $r=1,2$ si ha che

$$f\left(\frac{1}{\sqrt2},\frac{1}{\sqrt2}\right)=f(\sqrt2,\sqrt2)=0$$
$$f\left(\frac{1}{\sqrt2},-\frac{1}{\sqrt2}\right)=2,\ f(\sqrt2,-\sqrt2)=8$$

Confrontando i valori trovati si ha che (x,x) per $x\in\left[\frac{1}{\sqrt2},\sqrt2\right]$ sono punti di minimo assoluto con valore 0 e $(\sqrt2,-\sqrt2)$ è un punto di massimo assoluto con valore 8.

5.f $f(x,y)=\dfrac{xy}{1+x^2+y^2}$ Max/min su $D=\{(x,y): x^2+y^2\leq 1\}$?

$f\in C^2(\mathbb{R}^2)$, D è compatto.



$$\partial D=\Gamma=\{(x,y): x^2+y^2=1\}$$
$$=\{(\cos\theta,\sin\theta): \theta\in[0,2\pi)\}$$

Punti stazionari interni a D:

$$\nabla f(x,y)=\left(\frac{y(1+x^2+y^2)-2x^2y}{(1+x^2+y^2)^2}, \frac{x(1+x^2+y^2)-2xy^2}{(1+x^2+y^2)^2}\right)=(0,0)$$

$$\begin{cases} y(1-x^2+y^2)=0 \\ x(1+x^2-y^2)=0 \end{cases} \rightarrow y=0,\ y^2=x^2-1$$

$$\begin{cases} y=0 \\ x(1+x^2)=0 \end{cases} \cup \begin{cases} y^2=x^2-1 \rightarrow y^2=-1 \text{ impossibile} \\ x\cdot 2=0 \rightarrow x=0 \end{cases}$$

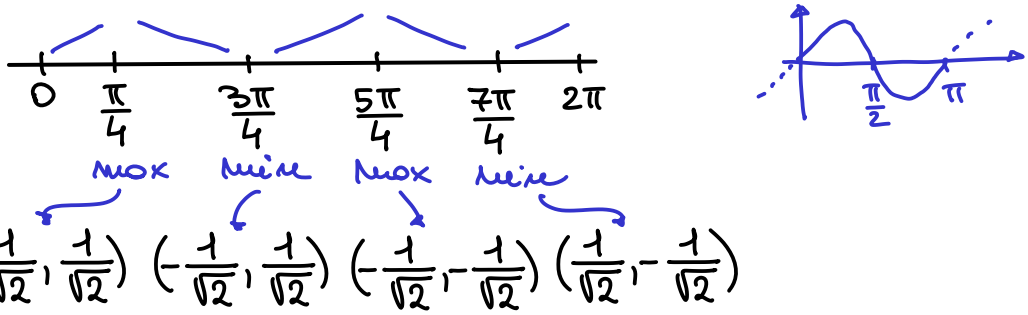
$\hookrightarrow x=0$

L'unico punto stazionario è $(0,0)$ ed è interno a D.

$$\boxed{f(0,0)=0}$$

Bordo di D: consideriamo per $\theta \in [0,2\pi)$

$$h(\theta)=f(\cos\theta,\operatorname{sen}\theta)=\frac{1}{2}\cos\theta\cdot\operatorname{sen}\theta=\frac{1}{4}\operatorname{sen}(2\theta)$$



$$\left(\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right)\ \left(-\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right)\ \left(-\frac{1}{\sqrt{2}},-\frac{1}{\sqrt{2}}\right)\ \left(\frac{1}{\sqrt{2}},-\frac{1}{\sqrt{2}}\right)$$

$$\boxed{f\left(\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right)=f\left(-\frac{1}{\sqrt{2}},-\frac{1}{\sqrt{2}}\right)=\frac{1}{4}} \qquad \boxed{f\left(-\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right)=f\left(\frac{1}{\sqrt{2}},-\frac{1}{\sqrt{2}}\right)=-\frac{1}{4}}$$

Confrontando i valori ottenuti si ha che:

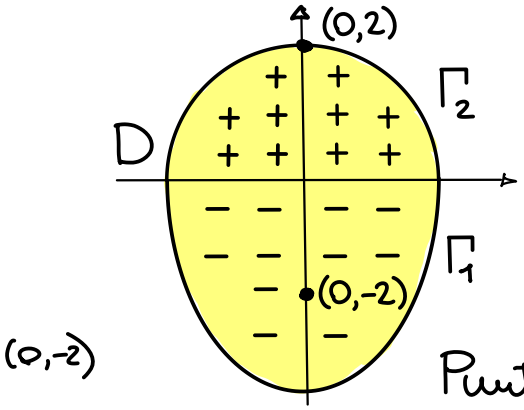
$\left(\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right)$ e $\left(-\frac{1}{\sqrt{2}},-\frac{1}{\sqrt{2}}\right)$ sono punti di massimo assoluto con valore $\frac{1}{4}$ e $\left(-\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right),\left(\frac{1}{\sqrt{2}},-\frac{1}{\sqrt{2}}\right)$ sono punti di minimo assoluto con valore $-\frac{1}{4}$.

$(x\ y,z)$

$(\cos t,\operatorname{sen} t)$
$\vec{\sigma}_x$

5.9 $f(x,y)=y(y-x^2+4)$ Max/min in $D=\{(x,y): x^2-4\le y\le\sqrt{4-x^2}\}$?

$f\in C^2(\mathbb{R}^2)$, D è compatto.



Bordo di D: $\partial D=\Gamma_1\cup\Gamma_2$

$\Gamma_1=\{(x,x^2-4): x\in[-2,2]\}$

$\Gamma_2=\{(x,y): x^2+y^2=4, y\in(0,2]\}$

Punti stazionari interni a D:

$$\nabla f(x,y)=(-2xy,2y-x^2+4)=(0,0)$$

$$\begin{cases}-2xy=0 \to x=0 \text{ oppure } y=0\\ 2y-x^2+4=0\end{cases} \iff \begin{cases}x=0\\ y=-2\end{cases} \quad \begin{cases}y=0\\ x=\pm 2\end{cases}$$

$(0,-2)$ $\quad$ $(\pm 2,0)$

Solo $(0,-2)$ è interno a D e $\boxed{f(0,-2)=-4}$

Bordo di D:

1) In Γ_1 la funzione vale $\boxed{f(x,x^2-4)=0}$

2) Per Γ_2 usiamo i moltiplicatori con $g(x,y)=x^2+y^2-4=0$.

$$\begin{cases}-2xy=\lambda\cdot 2x \to x(\lambda+y)=0 \text{ oppure } x=0 \text{ oppure } \lambda=-y\\ 2y-x^2+4=\lambda\cdot 2y\\ x^2+y^2=4\end{cases}$$

$$\iff \begin{cases}x=0\\ 2y+4=\lambda 2y\\ y^2=4\end{cases} \cup \begin{cases}\lambda=-y\\ 2y-x^2+4=-2y^2\\ x^2+y^2=4\end{cases}$$

$y^2=4 \to y=\pm 2$

$2y+y^2=-2y^2 \to y=0$ oppure $y=-\frac{2}{3}$

$$\begin{cases} x=0 \\ y=2 \\ \lambda=2 \end{cases} \cup \begin{cases} x=0 \\ y=-2 \\ \lambda=0 \end{cases} \cup \begin{cases} y=0 \\ x=\pm 2 \\ \lambda=0 \end{cases} \cup \begin{cases} y=-\frac{2}{3} \\ x=\pm\frac{4\sqrt{2}}{3} \\ \lambda=\frac{2}{3} \end{cases}$$

$(0,2)\in\Gamma_2$ $\quad (0,-2)\notin\Gamma_2$ $\quad (\pm 2,0)\notin\Gamma_2$ $\quad \left(\pm\frac{4\sqrt{2}}{3},-\frac{2}{3}\right)\notin\Gamma_2$

$(0,2)$ è l'unico punto stazionario vincolato e

$$\boxed{f(0,2)=12}$$

Confrontando i valori, $(0,2)$ è un punto di massimo assoluto con valore 12 e $(0,-2)$ è un punto di minimo con valore -4.

5.h $\quad f(x,y)=x^2+y^2-13x$

Max/min in $D=\{(x,y): |x-2|\geq 2+|y^2-4|, 0<x\leq 8\}$?

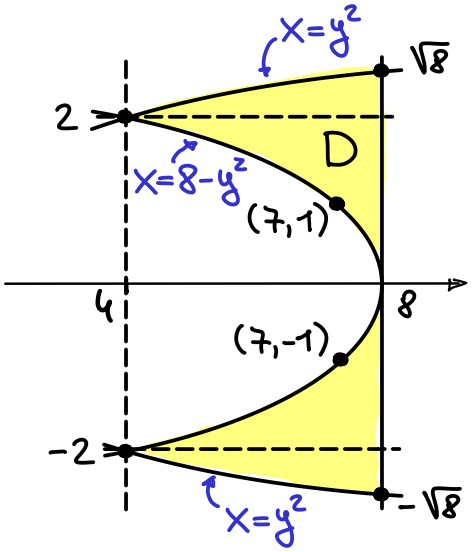
$f\in C^2(\mathbb{R}^2)$, D è compatto. Se $0<x\leq 8$ allora

$|x-2|\geq 2+|y^2-4|$ diventa:

- $2-x\geq 2+|y^2-4|$ se $0<x<2$ (impossibile: $2-x<0$, $|y^2-4|\geq 0$)
- $x-2\geq 2+|y^2-4|$ se $2\leq x\leq 8$

$x\geq 4+|y^2-4|$ diventa:

- $x\geq 4+y^2-4$ se $|y|\geq 2$
- $x\geq 4+4-y^2$ se $|y|\leq 2$ $\quad (8-y^2)$



Bordo di D: $\partial D=\Gamma_1\cup\Gamma_2\cup\Gamma_3$

$\Gamma_1=\{(x,y): g_1(x,y)=x+y^2-8=0, x\in[4,8]\}$

$\Gamma_2=\{(x,y): g_2(x,y)=y^2-x=0, x\in[4,8]\}$

$\Gamma_3=\{(8,y): y\in[-\sqrt{8},\sqrt{8}]\}$

∂D è regolare a tratti.

$\nabla f(x,y) = (2x-13, 2y)$ è l'unico punto stazionario $\left(\frac{13}{2}, 0\right) \notin D$. Dato che D è compatto e f è continua i punti di max/min assoluto stanno in ∂D.

Γ_1) Moltiplicatori di Lagrange:

$$\begin{cases} 2x-13 = \lambda \cdot 1 \\ 2y = \lambda 2y \\ x+y^2 = 8 \\ 4 \le x \le 8 \end{cases} \rightarrow \lambda = 1 \vee y = 0$$

$$\begin{cases} \lambda = 1 \\ x = 7 \\ y = \pm 1 \end{cases} \cup \begin{cases} y = 0 \\ x = 8 \end{cases}$$

$\boxed{f(7, \pm 1) = -41}$ $\boxed{f(8,0) = -40}$

Inoltre agli estremi $\boxed{f(4, \pm 2) = -32}$

Γ_2) Moltiplicatori di Lagrange:

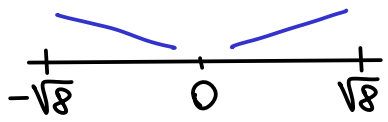
$$\begin{cases} 2x-13 = -\lambda \cdot 1 \\ 2y = \lambda 2y \\ y^2 = x \\ 4 \le x \le 8 \end{cases} \rightarrow \lambda = 1 \vee y = 0$$

$$\begin{cases} \lambda = 1 \\ x = 6 \\ y = \pm\sqrt{6} \end{cases} \cup \begin{cases} y = 0 \\ x = 0 \end{cases} \notin \Gamma_2$$

$\boxed{f(6, \pm\sqrt{6}) = -36}$

Inoltre agli estremi $\boxed{f(4, \pm 2) = -32, f(8, \pm\sqrt{8}) = -32}$

Γ_3) Restrizione $f(8,y) = -40 + y^2$



I punti $(8, \pm\sqrt{8})$ e $(8,0)$ sono già stati considerati.

Confrontando i valori, $(4, \pm 2)$ e $(8, \pm\sqrt{8})$ sono punti di massimo assoluto con valore -32 e $(7, \pm 1)$ sono punti di minimo assoluto con valore -41.